

OpenTelemetry Browser

what exists, what's coming, and how we can work together

<https://github.com/open-telemetry/community#sig-browser>

<https://github.com/open-telemetry/opentelemetry-browser>

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What is OpenTelemetry?

- Vendor-neutral standard for generating and shipping telemetry
- APIs + SDKs + a wire protocol (OTLP, the OpenTelemetry Protocol) + shared vocabulary
- Not a backend, not a dashboard, no storage

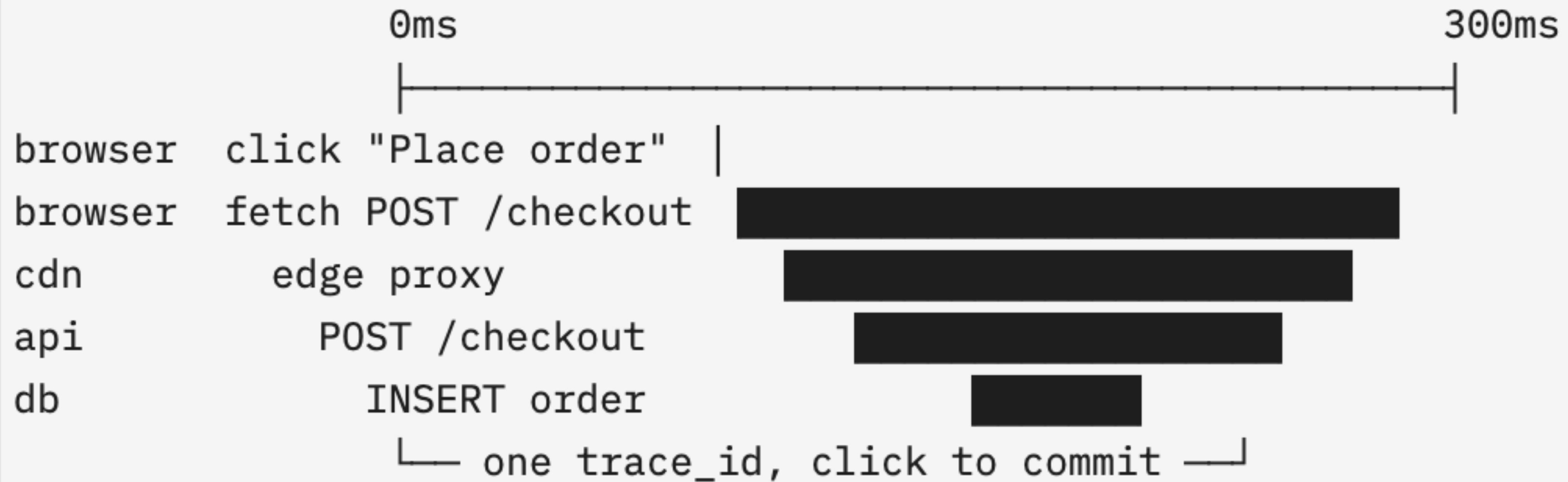
Who is OpenTelemetry?

- Cloud Native Computing Foundation
- CNCF is the vendor-neutral hub of cloud native computing, dedicated to making cloud native ubiquitous
- OTel is consistently ranked second-most-active after Kubernetes
- Most of the major observability vendors contribute
- Specs + a dozen language implementations + semantic conventions

Signals

- Traces: tree of timed operations (spans) across services
- Metrics: aggregated numbers over time
- Logs: timestamped records; a log record describes a discrete, named occurrence
- RUM translation: beacon $\approx$ log, waterfall $\approx$ trace
- In clients (web and mobile) we use spans + logs; metrics are derived later in the pipeline, not emitted from the client (aggregation belongs server-side: cardinality, CPU, battery)

The gap



How the pieces fit

- app code → API (no-op safe) → SDK → OTLP → Collector → any backend
- API/SDK split: libraries instrument once against a stable API
- Collector: transform, scrub, sample, route off the client
- Portability: switch backends without re-instrumenting

Semantic Conventions

- Registry of agreed attribute names: `http.request.method`, `url.full`, `browser.document.url.full`, ...
- Works like a standards body: proposals, review, stability levels
- Proof the bridge works: Trace Context (traceparent) is a W3C Recommendation

The browser today

- Dedicated OTel SIG (Special Interest Group) for the browser runtime
- Scope today: instrumentations, data model, convention proposals
- Ships today: @opentelemetry/browser-instrumentation (experimental, v0.5.x)
- No browser-tailored SDK yet: today you pair these instrumentations with the general JS SDK pieces; the repo is the SDK's future home

Built on this group's work

- PerformanceObserver, Navigation Timing, Resource Timing
- Navigation API for soft navigations (opt-in; History patching fallback)
- requestIdleCallback
- OTel consumes what W3C ships; better platform APIs → better telemetry

Log record

```
{
  "eventName": "browser.web_vital",
  "timestamp": 1718040000123,
  "severityNumber": 9,
  "attributes": {
    "browser.web_vital.name": "lcp",
    "browser.web_vital.value": 1840.2,
    "browser.web_vital.delta": 1840.2,
    "browser.web_vital.rating": "good",
    "browser.web_vital.id": "v5-1718040000123-882471",
    "browser.web_vital.navigation_type": "navigate"
  },
  "resource": { "service.name": "shop-frontend", " ... ": " ... " },
  "traceId": "4bf92f3577b34da6a3ce929d0e0e4736",
  "spanId": "00f067aa0ba902b7"
}
```

How the SIGs fit together

- Browser SIG: instrumentations + data model for the browser runtime
- Semantic Conventions SIG: separate group, owns the attribute registry; the browser.* conventions land there (overlap, not exactly the same people)
- Cross-SIG collaboration is routine: client-side exceptions, clicks/taps, and sessions are shared ground with mobile

Get involved

- Weekly call: Thursdays 8:30 PT (public notes, open to anyone)
- CNCF Slack: #otel-browser
- GitHub: [open-telemetry/opentelemetry-browser](https://github.com/open-telemetry/opentelemetry-browser) + semantic convention PRs
- No code required: reviewing a convention definition is a contribution

Shared development

- RUM CG → OTel: comment on the open browser.* convention PRs (they live in the Semantic Conventions SIG; we'll point you at the right ones)
- OTel → RUM CG: we'll bring our platform pain points:
- Reliable telemetry delivery at page exit
- Soft navigation detection
- Web vitals for soft navigations
- More to come

Questions?